

- 18 C** Note that the vertical nylon lines result in the horizontal diffraction images while the horizontal nylon lines results in the vertical diffraction images.

From the equation for diffraction grating, $d\sin\theta = n\lambda$

For monochromatic light of fixed λ , For the same order of maxima, when d is larger, $\sin\theta$ is smaller and hence θ is smaller. This means that the separation of the straight through image and the image of the particular order is closer together.

Since the separation of the straight through image and the vertical images are smaller compared to the separation of the straight through image and the horizontal images, the d (slit separation) for the horizontal lines must be larger than the d for the vertical lines.